Case Study

Airport Congestion & Slot Utilization Optimization

(Aviation Industry Case Study)

1. Company Overview

Runewind Aviation, based in the Netherlands, is a fast-growing airport operations and air traffic optimization company responsible for managing flight coordination, runway slot allocation, and ground traffic efficiency across multiple **European international airports**.

History & Growth Journey

Runewind Aviation was founded in 2012 as part of a European aviation modernization initiative aligned with regional air traffic harmonization efforts.

Key milestones include:

- **2015:** Introduction of semi-automated flight scheduling systems aligned with European slot coordination standards
- **2018:** Expansion into three major EU international airports
- **2021:** Adoption of cloud-based data infrastructure integrated with regional air traffic systems
- **2024:** Launch of digital slot allocation and congestion monitoring system compliant with EU aviation frameworks

Market Presence

Runewind Aviation currently manages operational coordination for:

- 6 major international airports across Europe
- Over 1,500 daily flight movements
- Multiple EU-based and international airline partners

Core Services

- Airport slot allocation and scheduling
- Runway and gate utilization management
- Airside congestion monitoring
- Coordination with airlines, airport authorities, and regional air traffic control systems

Differentiators

- Expertise in high-density, highly regulated European airspace
- Integration with centralized European air traffic systems like Eurocontrol
- Real-time cross-border operational coordination
- Strong compliance with EU slot allocation regulations

2. Business Challenge

Runewind Aviation faces a critical operational bottleneck: inefficient airport slot utilization and congestion during peak travel periods—particularly in high-traffic hubs.

Current Situation

- Major European hubs (e.g., Amsterdam Airport Schiphol) operate at or near capacity during peak hours

- Delays are increasingly driven by slot misalignment rather than technical faults
- Airlines frequently request dynamic slot adjustments due to network-wide delays
- Ground congestion impacts turnaround efficiency

Core Pain Points

1. **Inefficient Slot Allocation**
   Reliance on historical scheduling patterns rather than predictive demand modeling
2. **Limited Real-Time Congestion Visibility**
   Lack of unified dashboards across airports and control systems
3. **Demand Volatility**
   Driven by:
   - Weather disruptions
   - Seasonal tourism spikes
   - Cross-border airline dependencies
4. **Cascading Delay Effects**
   One delay impacts multiple airports due to tightly interconnected European routes
5. **Fragmented Data Systems**
   Data spread across airline systems, airport databases, and regional ATC platforms

Business Impact

- Increased flight delays
- Reduced throughput at already capacity-constrained airports
- Airline dissatisfaction and regulatory pressure
- Passenger experience degradation

3. Rationale for the Project

What is Airport Slot Utilization Optimization?

A data-driven approach to allocating takeoff and landing slots that maximizes runway efficiency while complying with strict European aviation regulations.

Real-World Context

Major European hubs such as:

- Heathrow Airport
- Amsterdam Airport Schiphol
- Frankfurt Airport

use advanced slot coordination systems under EU regulations, yet still face congestion due to high demand and limited expansion capacity.

Why This Project is Strategically Important

1. Maximize Existing Infrastructure (Europe has limited space for expansion)
2. Reduce Delays in High-Traffic Corridors
3. Improve Airline Network Efficiency Across Borders
4. Enable Real-Time, Data-Driven Decisions
5. Enhance Passenger Experience in Major Hubs

4. Project Objectives

The primary goal of this project is to design and implement a data-driven airport slot optimization system that improves operational efficiency and reduces congestion.

Key Objectives

1. Build a Centralized Airport Data Platform

- Consolidate flight schedules, runway usage, and airline data into Snowflake
- Ensure a single source of truth for all airport operations

2. Develop Predictive Congestion Models

- Use Python-based machine learning models to forecast congestion levels
- Predict peak congestion windows 24–72 hours in advance

3. Optimize Slot Allocation Logic

- Implement linear programming models to assign optimal runway slots
- Reduce overlapping and inefficient scheduling

4. Create Real-Time Operational Dashboards

- Use Power BI to visualize:
    - Runway utilization
    - Gate occupancy
    - Delay heatmaps

5. Improve Operational Coordination Workflow

- Use Jira to manage optimization backlog and system improvements
- Use Confluence to document operational rules and scheduling policies

6. Deploy Decision Support Interface

- Build a Gradio-based tool for:
    - Slot simulation
    - Congestion prediction input
    - “What-if” operational scenarios

5. Comprehensive Dataset Design

Table 1: Flights

- Columns:
  - flight_id
  - airline_id
  - origin_airport
  - destination_airport
  - scheduled_departure_time
  - actual_departure_time
  - delay_minutes
- Relationships:
  - Linked to Airports and Airlines tables

Table 2: Airports

- Columns:
  - airport_id
  - airport_name
  - location
  - runway_capacity_per_hour
  - operational_hours

Table 3: Runway_Usage

- Columns:
  - runway_id
  - airport_id
  - flight_id
  - usage_start_time

    - usage_end_time
    - status (scheduled / delayed / completed)
- Relationships:
    - Many-to-one with Airports
    - Many-to-one with Flights

Table 4: Slot_Allocation

- Columns:
    - slot_id
    - flight_id
    - airport_id
    - assigned_time
    - slot_status (confirmed / delayed / rescheduled)
- Relationships:
    - Central table connecting Flights and Airports

Table 5: Weather_Data

- Columns:
    - weather_id
    - airport_id
    - timestamp
    - visibility
    - wind_speed
    - precipitation_level
- Relationships:
    - Linked to Airports and Flights

Table 6: Airlines

- Columns:
    - airline_id
    - airline_name
    - country
    - fleet_size
- Relationships:
    - Connected to Flights table

Data Model Overview

- Flights is the central transactional table
- Airports acts as a master dimension table
- Runway_Usage and Slot_Allocation form operational fact tables
- Weather_Data enriches predictive modeling
- Relationships are structured in a star schema optimized in Snowflake

6. Technology Stack

Data Infrastructure

- Snowflake → centralized aviation data warehouse

Analytics & Visualization

- Power BI → congestion dashboards & operational KPIs

Data Science & Optimization

- Python → machine learning & linear programming
- Libraries: Pandas, Scikit-learn, PuLP

Project Management

- Google docs, Google Sheets, Google drive, Google Slides

Decision Interface

- Gradio → interactive slot simulation & congestion prediction tool

7. Project Implementation Scope (End-to-End Workflow)

Phase 1: Project Initiation (Project Management – Jira & Confluence)

- Define project scope and stakeholders
- Create epics in Jira:
    - Data pipeline setup
    - ML model development
    - Dashboard creation
- Document requirements in Confluence
- Define KPIs (delay reduction, slot efficiency, congestion score)

Phase 2: Data Engineering (Snowflake)

- Ingest flight, runway, and weather datasets into Snowflake
- Build ETL pipelines for:
    - Flight schedules
    - Slot allocations
    - Historical congestion patterns
- Clean and normalize data into star schema

Phase 3: Data Analysis (Power BI)

- Build operational dashboards:
  - Runway utilization heatmaps
  - Delay trend analysis
  - Peak congestion visualization
- Identify bottlenecks and inefficiencies

Phase 4: Data Science & Optimization (Python)

- Develop:
  - Congestion prediction model (classification/regression)
  - Slot optimization using linear programming
- Train models using historical flight + weather data
- Evaluate performance using delay reduction metrics

Phase 5: Decision Support System (Gradio)

- Build interactive tool for:
  - Simulating slot allocation changes
  - Predicting congestion under different scenarios
- Enable airport planners to test “what-if” decisions

Phase 6: Monitoring & Continuous Improvement

- Track system performance via Power BI dashboards
- Use Jira to log improvement requests
- Update Confluence with new operational learnings
- Retrain ML models periodically with new data

Final Expected Outcomes

- Improvement in slot utilization efficiency
- Significant reduction in runway congestion
- Improved on-time departure performance
- Data-driven airport operations culture
- Faster operational decision-making

One week breakdown - DA, DS and DE

1-WEEK EXECUTION PLAN

Airport Congestion & Slot Utilization Optimization System

Overall Sprint Structure

Day	Focus
Day 1	Setup + Requirements + Data Audit
Day 2	Data Engineering (Snowflake ingestion)
Day 3	Data Modeling + Data Quality + BI foundation
Day 4	Data Analysis + Dashboard building
Day 5	Data Science modeling (prediction + optimization)
Day 6	Gradio app + integration + PM/BA validation
Day 7	Final testing + documentation + presentation

ROLE BREAKDOWN (PARALLEL WORKSTREAMS)

DAY 1 — PROJECT INITIATION (ALL TEAMS ALIGN)

Data Engineer (DE)

- Inspect CSV files:
  - Flights.csv
  - Slots.csv
  - Runway_usage.csv
  - Weather.csv
- Define schema mapping for Snowflake
- Identify missing / null data issues

Data Analyst (DA)

- Perform quick EDA in Python:
  - Missing values
  - Delay distribution
  - Peak hours analysis
- Create initial insights draft

Data Scientist (DS)

- Define modeling approach:
  - Congestion classification model
  - Slot optimization approach (linear programming)
- Identify features required

DAY 2 — DATA ENGINEERING (SNOWFLAKE BUILD)

Data Engineer (PRIMARY OWNER DAY 2)

Step 1: Snowflake Setup

- Create database:
  - AVIATION_DB
- Create schemas:
  - RAW
  - PROCESSED

Step 2: Load CSVs into staging tables

- Use Snowflake stages:
  - Internal stage for CSV upload
- Load tables:
  - FLIGHTS_RAW
  - SLOTS_RAW
  - RUNWAY_RAW
  - WEATHER_RAW

Step 3: Data Cleaning SQL transformations

- Standardize:
  - timestamps
  - airport codes
- Remove duplicates
- Handle null delays

Output of Day 2:

✔ Clean structured dataset in Snowflake
✔ Raw → processed pipeline working

DAY 3 — DATA MODEL + QUALITY + BI FOUNDATION

Data Engineer

- Build star schema:

Fact Tables:

- flight_operations
- runway_usage
- slot_allocation

Dimension Tables:

- airports
- airlines
- weather

Data Analyst

- Connect Power BI to Snowflake
- Build dataset connections

Start dashboards:

- Delay trends
- Airport congestion heatmap (basic)

DS

- Feature engineering plan:
    - hour_of_day
    - weather severity index
    - historical delay lag features

DAY 4 — DATA ANALYSIS + DASHBOARDS

Data Analyst (MAIN OWNER DAY 4)

Power BI dashboards:

1. Runway Utilization Dashboard
2. Slot Allocation Efficiency
3. Delay Heatmap (Airport vs Time)
4. Congestion Timeline

Insights to extract:

- Peak congestion hours
- Airports with highest inefficiency
- Correlation: weather vs delays

DE

- Optimize Snowflake queries for Power BI performance

DAY 5 — DATA SCIENCE MODELS

Data Scientist (MAIN OWNER DAY 5)

Model 1: Congestion Prediction

- Input:
  - time
  - airport
  - weather
  - historical load
- Output:
  - congestion risk score

Models:

- XGBoost / Random Forest

Model 2: Slot Optimization

- Use linear programming:
  - minimize idle runway time
  - maximize throughput

Library:

- PuLP / SciPy

Output:

✔ Prediction model trained
✔ Optimization engine built

DA

- Validate model outputs vs historical patterns

DAY 6 — GRADIO APP + INTEGRATION

DS + DE (JOINT)

Build Gradio App:

Features:

- Input:
  - airport
  - time window
  - weather conditions
- Output:
  - congestion probability
  - recommended slot adjustments

DA

- Test outputs visually

DAY 7 — FINALIZATION + PRESENTATION

DA

- Final Power BI dashboard polishing
- Add executive summary page

DS

- Validate model accuracy metrics
- Document ML pipeline

DE

- Final Snowflake optimization + cleanup

FINAL DELIVERABLES

Data Science

- Congestion prediction model
- Slot optimization engine
- Gradio decision tool

Data Analytics

- Executive Power BI dashboard
- Airport performance insights

Data Engineering

- Snowflake data warehouse
- ETL pipeline from CSV → structured model

One week breakdown - PM and BA

1-WEEK EXECUTION PLAN: Airport Congestion & Slot Optimization

Project Management & Business Analysis Focus

Role	Primary Deliverable(s)
Business Analyst (BA)	**Business Requirements Document (BRD)** (Iterative Daily Chapters)
Project Manager (PM)	**Project Charter, Stakeholder Register, Communication Plan, Risk Register**

Daily Deliverable Breakdown

Day 1: Foundation & Alignment

- **BA:** Drafts **Section 1 of BRD**: Project Scope, Executive Summary, and Current State Analysis (The "Problem" of congestion).
- **PM:** Completes the **Project Charter** (Objectives, Milestones, Success Criteria) and the **Stakeholder Register** (Identifying Airport Ops, Airlines, and Tech Team).
- **Tech Sync:** Align on CSV field definitions vs. Business KPIs.

Day 2: Requirements & Governance

- **BA:** Drafts **Section 2 of BRD**: Data Requirements & Functional Requirements (User stories for Snowflake ingestion and data cleaning logic).

- **PM:** Completes the **Communication Plan** (Internal daily standup cadence + Stakeholder reporting frequency).
- **Tech Sync:** Validate that the Snowflake schema supports the BA's defined KPIs.

Day 3: Logic & Risk Mapping

- **BA:** Drafts **Section 3 of BRD**: Business Logic & KPI Definitions (Defining "Congestion Thresholds" and "Slot Utilization" formulas).
- **PM:** Completes the **Risk Register** (Identifies risks like data latency, model inaccuracy, and Snowflake credit overages).
- **Tech Sync:** Ensure the Star Schema (Fact/Dim tables) mirrors the BA's logic.

Day 4: Analytics & Refinement

- **BA:** Drafts **Section 4 of BRD**: Reporting & UI Requirements (Wireframes for the Power BI dashboards and visualization logic).
- **PM:** Risk Mitigation—Review the Risk Register against Day 4 progress; update Stakeholders on any timeline deviations.
- **Tech Sync:** Review Power BI drafts to ensure they meet the BA's "Decision Support" criteria.

Day 5: Predictive Logic & Constraint Validation

- **BA:** Drafts **Section 5 of BRD**: Model Specifications (Non-functional requirements for the Optimization engine and Prediction accuracy targets).
- **PM:** Mid-sprint health check—Update the Project Charter status and ensure DS team has no blockers for the PuLP/SciPy integration.
- **Tech Sync:** Validate that the ML model features align with business constraints.

Day 6: Integration & UAT Prep

- **BA:** Finalizes **Full BRD**: Combines all sections, adds Glossary and Traceability Matrix.
- **PM:** Finalizes the **Stakeholder Register & Communication Plan** for the Handover/Demo phase (Who gets the final report?).
- **Tech Sync:** PM/BA validation of the Gradio App. Does the "Recommended Slot Adjustment" make business sense?

Day 7: Delivery & Closure

- **BA: Final Deliverable:** Comprehensive 1-Week BRD.
- **PM: Final Deliverable:** PM Package (Charter, Stakeholder Register, Comm Plan, Risk Register) + Project Post-Mortem summary.
- **Final Presentation:** Joint walkthrough of the System (Snowflake + Power BI + Gradio) to stakeholders.